Program 5.1 Find the sum of all the primes below two million.

Solution:

```
def isPrime(n):
    if n < 2: return "Neither prime, nor composite"
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True
sum = 0
for i in range(2, 2000000):
    if isPrime(i):
        sum += i
print (sum)
```

Program 5.2 By considering the terms in the Fibonacci sequence whose values do not exceed four million, WAP to find the sum of the even-valued terms.

Solution:

```
prev, cur = 0, 1
total = 0
while True:
    prev, cur = cur, prev + cur
    if cur >= 4000000:
        break
    if cur % 2 == 0:
        total += cur
print(total)
```

VIVA QUESTIONS:

1. What are default arguments?
2. What are variable length arguments?
3. What are keyword arguments?
4. Mention the use of map().
5. Mention the use of filter().
6. Mention the use of reduce().
7. Define a string.
8. How is string slicing done?
9. What is the usage of repetition operator?
10. How is string concatenation done using + operator>
11. Mention some string methods.
12. How is length of a string found?
13. How is a string converted to its upper case?
14. Differentiate isalpha () and isdigit().

Program 6.1 : Write a program to count the numbers of characters in the string and store them in a dictionary data structure.
Solution :

Method 1:

```
mystring = "my name is nitin"
all_freq = {}
for i in mystring:
 if i in all_freq:
   all_freq[i] += 1
 else:
   all_freq[i] = 1
print(all_freq)
```

Method 2:

```
from collections import Counter
mystring = "my name is nitin"
freq = Counter(mystring)
print ( freq)
```

Program 6.2 Write a program to use split and join methods in the string and trace a birthday of a person with a dictionary data structure
Solution :

```
mystring = "my name is nitin"
split_string=mystring.split(' ')
print(split_string)
join_string='-'.join(split_string)
print(join_string)
```

```
//*********Trace a Birthday of a person with a dictionary data structure*******///
birthdays ={'nitin':'25/03/1988','nishant':'01/09/1990'}
name=input('enter the name of person')
if name in birthdays:
    print(birthdays[name])
#print('{} birthday date is {}'.format(name,birthdays[name]))
else:
    print("Not match the name in dictionary")
```

VIVA QUESTIONS:

1. What is the usage of pop() in list?
2. Define a tuple.
3. Are tuples mutable or immutable?
4. Mention the use of return statement.
5. What is a Boolean function?
6. How is main function defined?
7. What is a dictionary?
8. How are tuples created?
9. How is a dictionary created?
10. How to print the keys of a dictionary?
11. How to print the values of a dictionary?
12. How is del statement used?
13. Can tuple elements be deleted?
14. What is Python interpreter?